Available online at www.sciencedirect.com

ScienceDirect

Procedia CIRP 109 (2022) 431–436

www.elsevier.com/locate/procedia

32nd CIRP Design Conference

Requirements analysis for an intelligent workforce planning system: a socio-technical approach to design AI-based systems

Stefan Gabriel[a*], Dominik Bentler[b], Eva-Maria Grote[a], Caroline Junker[a], David Meyer zu Wendischhoff[a], Michael Bansmann[c], Benedikt Latos[c], Daniela Hobscheidt[a], Arno Kühn[a], Roman Dumitrescu[d]

[a]*Fraunhofer Institute for Mechatronic Systems Design IEM, Zukunftsmeile 1, 33102 Paderborn, Germany*
[b]*University of Bielefeld, Universitätsstraße 25, 33615 Bielefeld*
[c]*Miele & Cie. KG, Carl-Miele-Straße 29, 33332 Gütersloh*
[d]*Heinz Nixdorf Institute University of Paderborn, Fürstenallee 11, 33102 Paderborn, Germany*

* Corresponding author. Tel.: +495251-5465-114; fax: +495251-5465-102. *E-mail address:* stefan.gabriel@iem.fraunhofer.de

Topic: Design methodology, methods and tools

Abstract

Artificial Intelligence (AI)-based applications promise great potential benefits for companies. However, an isolated consideration of the technical system is not sufficient for the design. Rather, it is necessary to design the entire work system taking into account the socio-technical system approach. This enables the combination of the strengths of people and intelligent systems. This paper presents an approach for a socio-technical requirements elicitation in the design of AI-based systems by adapting the HTO-analysis. First, a mission statement is developed. Based on a detailed process modelling, existing data and systems are recorded. In addition, all relevant stakeholder groups are included by conducting interviews and surveys. Thus, the procedure enables the derivation of a comprehensive catalogue of requirements. The application of the approach is illustrated by using an example from industrial practice, the design of an intelligent workforce planning system.

© 2022 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 32nd CIRP Design Conference

Keywords: socio-technial design; requirements elicitation; AI-human-collaboration; work design; workforce planning

1. Motivation

Artificial Intelligence (AI)-based applications offer great value for companies. For example, an AI-supported evaluation of existing large amounts of data can improve the quality of decisions and shorten processing times. At the same time, there are still complex challenges such as data protection, different data formats, lack of competences or poorly defined processes that make it difficult to tap the potential. To exploit the potentials, a systematic approach and implementation with close involvement of the affected employees is required [1], [2].

This article deals with the design of AI-based applications using the practical example of an intelligent workforce planning system in the assembly area of a manufacturing company. The focus of this paper is on the elicitation of requirements for AI-based systems by adapting the HTO analysis (**h**uman, **t**echnology, **o**rganization) according to ULICH (2013) [3].

The paper is structured as follows: after the problem analysis (section 2), current approaches in the examined topics are presented (section 3). Section 4 illustrates the used research methodology. Section 5 represents the main contribution of this paper. Here, the procedure for

2212-8271 © 2022 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the 32nd CIRP Design Conference
10.1016/j.procir.2022.05.274

This is a resupply of March 2023 as the template used in the publication of the original article contained errors. The content of the article has remained unaffected.

requirements elicitation and the application of this procedure is described. Finally, a conclusion is drawn in section 6.

2. Analysis of the Problem

There is no generally accepted definition of the term artificial intelligence. In this paper, the term AI refers to systems that exhibit behaviour commonly associated with human intelligence [1]. The terms AI-based system and intelligent system are used as synonyms in this paper.

AI will increasingly permeate processes in organizations. Fears exist, that AI will replace employees. However, only collaboration between humans and AI brings the best results. For example, as shown in Table 1, the complement of the strengths of humans and AI leads to the best possible decision-making processes [4], [5], [6].

Table 1. Complementary of humans and AI in decision-making situations according to JARRAHI (2018) [4]

	Human	AI
Uncertainty	Make swift, intuitive decisions in the face of the unkown	Provide access to realtime information
Complexity	Decide where to seek and gather data	Collect, curate, process and analyze data
Equivocality	Negotiate, build consensus, and rally support	Analyze sentiments and represent diverse interpretations

This paper pursues the idea of intelligence augmentation, i.e. the design of AI-based systems to support, but not to replace employees [4]. The aim is to design human-machine interaction in such a way, that mutual learning between humans and AI is possible and given [7]. This enables the combination of the strengths of humans and AI to optimize decision-making processes. For example, this can optimize complex planning and scheduling problems in manufacturing [8].

For this goal, a purely technology-centred approach for development does not seem sufficient. Instead, a human-centred approach seems promising in order to consider human action in addition to technical aspects. Thus, an early involvement of employees in the development of AI-based systems is required to identify the needs of the employees and to formulate corresponding requirements. For example, the human-machine interaction can be designed in accordance with the wishes of the employees. Furthermore, there is undocumented, implicit knowledge that can only be captured and considered by involving the relevant employees at an early stage [6], [9], [10].

This human-centred approach is closely linked to the socio-technical system approach. Here, a work system is described by linking social and technical subsystems. The HTO concept, illustrated in Fig. 1, highlights that a task is always performed by the interaction of human, technology and organization (also called MTO – man, technology, organization). Thus, an AI-based system (as the technical subsystem) must be designed and optimized in consideration of organizational structures and humans [3].

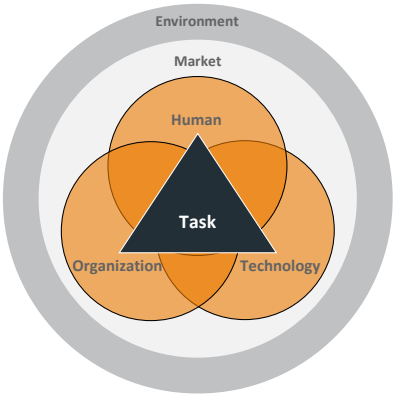

Fig. 1. HTO-Concept according to ULICH (2013) [3]

Based on the socio-technical system approach and the HTO concept, the HTO analysis has been developed as an integrated, holistic analysis tool. This analysis contains the different levels shown in Table 2 [3].

Table 2. Levels of the HTO-Analysis according to ULICH (2013) [3]

Level of HTO-Analysis	Description
Company	Corporate goals, strategy or organization as a frame of reference and boundary conditions
Organizational Unit	Examination of processes, forms of division of labour, etc.
Group	Identifying key activities and opportunities for collective regulation
Individual	Subjective assessment of the work situation and working conditions

These are run through in the order from the "highest" level (company) to the "lowest" level (individual), whereby the results of the analyses of the higher levels are included in the analyses of the lower levels. By this analysis, insights can be gained for the optimization of the design of the work systems [3]. In addition to this, ISO 9241-210 describe the human-centred design of interactive systems. This development process is iterative and involves the users [11].

However, there are various obstacles in the development of software-based systems: unclear objectives, numerous interactions and a resulting high level of complexity, as well as communication problems between the participants (e.g. due to different backgrounds and knowledge) [12].

When developing AI-based systems, specific challenges must also be taken into account, such as the availability of the required large amounts of data, the domain knowledge needed to create models, or regulatory and ethical aspects [13]. Regarding the employees in particular, implicit knowledge needs to be captured for example [10]. Furthermore, the behavior of the AI-based system should be explainable for the employees [6]. In addition to this, many companies have not yet an AI-application in use, so there is a lack of experience [1]. Moreover, the successful use of an AI-application requires extensive changes to many business processes because of two reasons. First, cross-disciplinary teams are necessary to define holistic solutions when the AI-application is developed. Second, the decisions and learnings from the

This is a resupply of March 2023 as the template used in the publication of the original article contained errors. The content of the article has remained unaffected.

AI-application also often affects wide parts and different levels of the organization [7].

This paper focus on the requirements elicitation, which is as a part of requirements analysis an essential subtask in the product development process. There are several prerequisites and possible techniques for eliciting requirements, which are shown as an excerpt in Table 3 [12].

Table 3. Elicitation of requirements according to PARTSCH (2010) [12]

Activies in advance	Investigation techniques	Support techniques
Find goals	Creativity techniques	Modelling
Find stakeholders	Observation techniques	Structuring
Classify goals	Questioning techniques	Workshop
Define system scope	Feedback techniques	Card Sorting

A systematic approach is particularly important for the identification of requirements to gather all the necessary information and to overcome the obstacles mentioned above [12]. For these reasons, it is necessary to regard the requirements elicitation for the development of AI-based systems from a socio-technical perspective. This enables the consideration of specific challenges in the development of AI-based systems and the optimization of the socio-technical system and the work system in a holistic way. These relevant topics and the deduced question are shown in Fig. 2.

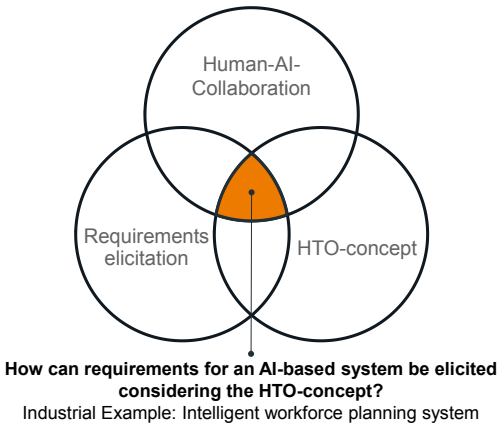

Fig. 2. Examined topics in this paper

In this paper, the development of an intelligent workforce planning system for the assembly area of a manufacturing company is considered as an example. In generalized form, the workforce planning process is described in Fig 3.

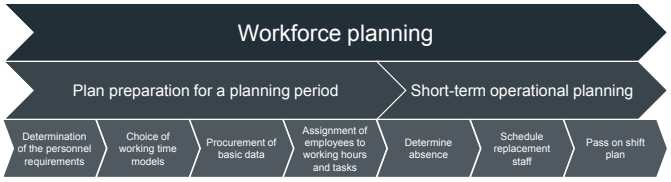

Fig. 3. Workforce planning process according to SCHERF (2005) [14]

Personnel planning in the assembly area of manufacturing companies is characterized by complex influencing factors and changes required at short notice. There are many influencing factors for the determination of required personnel capacity alone, which are listed in Table 4 [15].

Table 4. Excerpt of influencing factors for required personnel capacity according to BURGGRÄF ET AL. (2021) [15]

External Factors	Internal Factors
Market development / competition	Planned production programme
Demographic development	Planned production processes
Political development	Planned means of production
Social development	Work / company organisation
Tariff development	Absenteeism, fluctuation

Mastering this complexity through AI-based system support therefore holds great benefit potentials. In addition to mastering complexity, the aim is to achieve an improvement in work design through AI. Here, the support of the AI-based system for specifically controlling job rotation or team constellations is conceivable [16], [17]. Therefore, the assembly area is regarded as a work system. The primary work task is the assembly of products, but also secondary work tasks such as personnel planning are of particular importance for the work system as well [3]. Following the extended work system according to SCHLICK ET AL. (2018), the planning and effect system is also subdivided. Workforce planning can be classified as a task in the planning system, from which the executing work tasks (i.e. product assembly) in the effective system can be derived [18].

As already described, it is important to ensure a systematic approach to requirements elicitation in order to overcome the described challenges in the development of an AI-based system [12], [13]. The HTO analysis with its different levels of consideration offers a meaningful structuring in this context [3]. It ensures that the socio-technical perspective is taken into account, which is necessary due to the complex challenges of the intended cooperation between humans and AI-based systems. However, a complete HTO analysis is very comprehensive and time-consuming [19]. There is a need for an approach that adapts the HTO analysis for requirements elicitation for AI-based systems, considering constraints from (e.g.) the producing company or the project.

3. Current Methods

Various approaches in the literature deal with the socio-technical or human-centred design of technical systems. LTIFI ET AL. (2013) propose an approach to develop decision support systems from a human-centred perspective [20]. HUCHLER (2016) discusses development directions of Industry 4.0 in the context of the socio-technical systems approach, especially the human-technology relationship [9]. BANSMANN ET AL. (2019) propose a maturity-based approach for the introduction of digital technologies in work systems [21]. HOBSCHEIDT ET AL. (2020) develop socio-technical solutions pattern and an approach to use the pattern for individiual implementation paths [22]. PITZ ET AL. (2020) plead for the necessity of a participatory design of intelligent assistance systems and the consideration of the entire work system instead of an isolated view of the technical system [23]. These approaches demonstrate the relevance of socio-technical system design. However, they are insufficiently

This is a resupply of March 2023 as the template used in the publication of the original article contained errors. The content of the article has remained unaffected.

focused on the requirements elicitation of AI-based systems and their specific challenges.

VÖLKER ET AL. 2009 present a technical approach for a simulation-based workforce planning for complex assembly systems. The selection is made by algorithms that are defined in a rule catalogue and are selected or adapted problem-specifically [24]. SPATH ET AL. (2013) present an approach which involves employees in personnel planning by using a cyber-physical system [25]. RISCHMEYER (2021) describes the basic advantages, challenges and components of automated workforce planning in the healthcare sector [26]. These approaches show that the optimization of workforce planning by using digital technologies is seen as beneficial. However, no concrete methods or process models for the elicitation of requirements are discussed.

Summarizing all these aspects, it can be stated that various related approaches already exist in the literature, but socio-technical requirements elicitation in the context of AI-based systems has only been insufficiently considered so far.

4. Research Methodology

The need for an investigation of these aspects has become clear. A suitable research methodology is offered by the established Design Science Research methodology according to HEVNER (2007), which is shown in Fig. 4. This comprises the three areas environment, design and knowledge base as well as the relevance, design and rigour cycles [27].

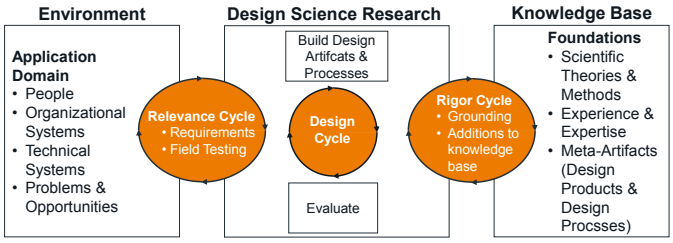

Fig. 4. Design Science Research Cycles according to HEVNER 2007 [27]

The procedure starts with the relevance cycle. In this cycle, the environment of the problem to be investigated is determined and relevant terms are defined. These can be, for example, people, technologies and organisations or companies. In this paper, the design of an intelligent workforce planning system in the assembly area of a manufacturing company is characterised as a problem. Furthermore, requirements for the research results are defined in this cycle. Within the rigor cycle, a link to the relevant knowledge base takes place, on which the research work is based. This consists of data as well as scientific theories and approaches, e.g. basics for personnel planning and requirements elicitation, methods for process modelling and surveys as well as existing standards. Within the design cycle, new findings and methods are developed that, on the one hand, meet the requirements of the relevance cycle and, on the other hand, build on the knowledge base of the rigour cycle. In addition, a continuous evaluation takes place [27].

The chosen research methodology offers a promising approach to systematically link requirements from the environment (including in combination with the socio-technical systems approach) with existing knowledge from the design of intelligent systems and, based on this, to develop new approaches and methods in the context of socio-technical requirements elicitation of AI-based systems.

5. Procedure Model & Application

The procedure presented in the following combines methods of requirements elicitation with elements of HTO analysis. Thus, a socio-technical requirements elicitation for an AI-based system is ensured. The procedure is applied to the presented example of an intelligent personnel planning system. Fig. 5 shows the four steps of the procedure model. Each step of the requirements elicitation aims at one level of the HTO analysis which are illustrated in Table 2. In each step, tried and tested methods are used. In order to ensure employee participation, technical experts, users as well as works councils were closely involved in the process.

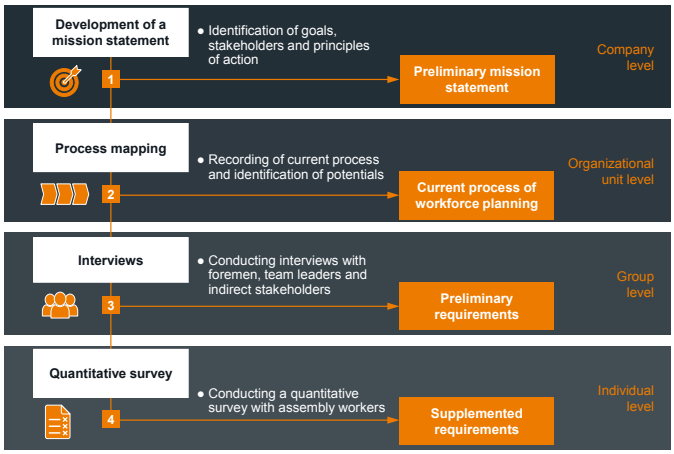

Fig. 5. Procedure for requirement elicitation

In **step 1**, a mission statement is developed according to GAUSEMEIER ET AL. (2019) [28]. The mission statement describes the desirable state and provides the basic direction. It serves as a framework for the design of an intelligent system. This is particularly important to develop a good understanding of the goals and impacts of AI-based systems among all participants, which is often missing due to a lack of experience with AI. The mission statement is related to intelligent workforce planning in the assembly area. However, indirectly involved company areas are also considered. Fig. 6 shows an excerpt of the mission statement.

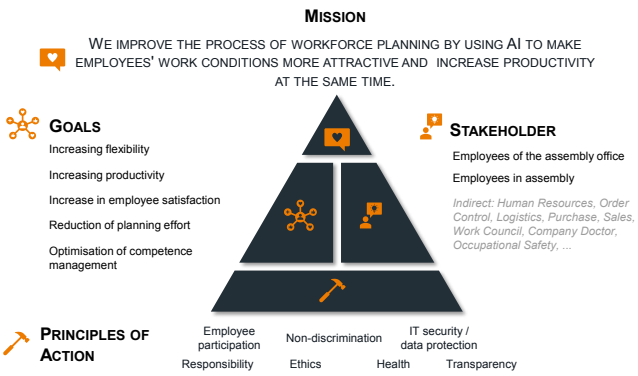

Fig. 6. Preliminary mission statement (excerpt)

This is a resupply of March 2023 as the template used in the publication of the original article contained errors. The content of the article has remained unaffected.

In this case, the two fundamental goals are an increased efficiency and effectiveness of the system as well as an improved well-being of the employees. These goals are based on ISO 6385 which describes the design of a work system [29]. The mission statement is not static and final, but preliminary. Dependent on the results of the following steps for the requirements elicitation, adjustments are possible.

In **step 2**, the personnel planning process was recorded using the OMEGA method (cf. GAUSEMEIER ET. AL. (2019)), which is shown schematically in Fig. 7 [28]. Here, personnel and capacity planning in assembly which are closely interrelated are considered. The process mapping serves to identify possible improvement potentials by AI and to be able to exactly specify the system support according to the individual process steps. For example, AI-potentials would be automated recommendations as to which employee could be deployed where, or the avoidance of media disruptions.

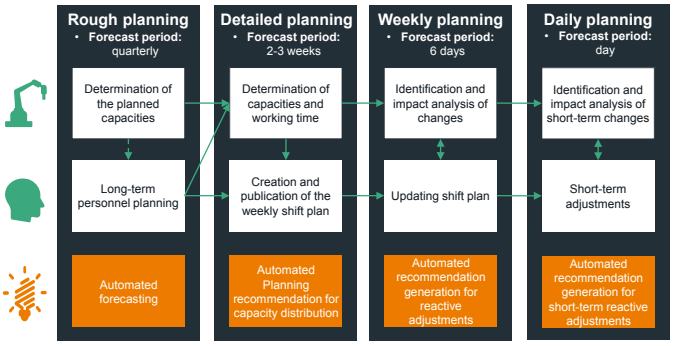

Fig. 7. Schematic process representation

In **step 3**, interviews were conducted to determine the actual requirements. Here, representatives of various stakeholder groups were interviewed about their expectations regarding an intelligent workforce planning system - expected main users of the tool (team leader, foreman, employees in the assembly office) as well as stakeholder of indirect areas (logistics, company doctor, ...). A total of 20 guided interviews were conducted, each lasting about 45 minutes. The interview statements were manually documented in writing. Based on GIOIA ET AL. (2012), categories are identified by means of inductive concept development [30]. The overview of these identified requirement categories is shown in Fig. 8.

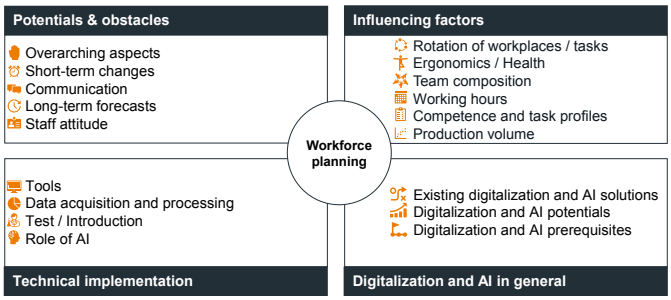

Fig. 8. Overview of the identified requirement categories

For example, various influencing factors such as working time models, competences or ergonomic could be identified. Currently, the workforce planning process is often strongly based on experience knowledge of the team leaders, which is insufficiently digitalized. Another result was that the participants wanted decision suggestions but not decision automation through AI.

In **step 4**, the work design, job satisfaction, work motivation and psychological well-being of the assembly workers which will be affected by the AI-based decisions were investigated by correlation analysis of a quantitative survey with 45 participants before the introduction of the technology. Fig. 9 shows an extract of the results (cf. BENTLER ET AL. (2022) [31] for a detailed description). This provided further insights into which aspects should be given special attention in the design of the AI-based system. For example, autonomy at shop floor level is rated an average of 2 on a 5-point scale. So, there is still a lot of scope to increase autonomy through an AI-based system if necessary.

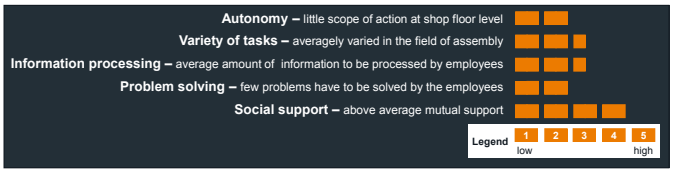

Fig. 9. Results of the quantitative survey (excerpt)

Once these steps were completed, the identified requirements were collated, reviewed, and prioritized. For this purpose, workshops have been held with the involvement of the different stakeholder groups. The result of this was a prioritized catalogue of 18 requirements for an intelligent tool for workforce planning. The requirements are divided in technical, employee-related and organizational categories in accordance with the idea of the HTO concept. An excerpt of this requirement catalogue is shown in Table 5.

Table 5. Requirement catalogue (excerpt)

Priority	Name
Very high	Recommendation generation for required short-term adjustments
Very high	Forecast of employee capacities
High	Automated shift plan creation
High	Consideration of team composition and team roles

6. Conclusion and Discussion

The single methods used in the approach are not new. However, the proposed approach combines proven methods of requirements elicitation with the HTO analysis. This has two main advantages. First, the requirements elicitation is conducted from a socio-technical perspective. This ensures the consideration of specific challenges of requirements elicitation for AI-based systems. Especially employee-related and organizational requirements are addressed and considered in this way. Second, the approach represents a simplification compared to the original HTO analysis, as it takes four instead of seven steps. This is an advantage for the application in industrial practice, as it saves time and capacity. In the related research project, this also ensured compliance with organizational restrictions and might be an advantage for

This is a resupply of March 2023 as the template used in the publication of the original article contained errors. The content of the article has remained unaffected.

other companies as well. The procedure is recommended for the requirements elicitation of AI-based systems, as it considers specific AI-challenges. However, it might be useful for the introduction of other technologies as well.

In the next step of the project, the iterative and incremental implementation with high involvement of the affected users is planned according to ISO 9241-210 [11]. This is particularly important because the requirements identified with this procedure are still at a high level of abstraction. The technical specifications still need to be derived from this and linked back to the requirements identified.

The procedure presented is illustrated in this article using the example of a single use case. Further research is needed to confirm that the proposed simplified form of HTO analysis is also useful for requirements elicitation for other use cases. Moreover, further research is needed to extend the procedure to the parallel introduction of several AI-based use cases. Here it is particularly necessary to consider the interactions between the use cases and to use possible synergies.

Acknowledgements

The results of this paper are based on the research project InTime within the Kompetenzzentrum Arbeitswelt.Plus. This research and development project is funded by the German Federal Ministry of Education and Research (BMBF) within "The Future of Value Creation – Research on Production, Services and Work" program and managed by the Project Management Agency Karlsruhe (PTKA). The author is responsible for the content of this publication.

References

[1] Bauer W, Ganz W, Hämmerle M, Renner T (Eds.). Künstliche Intelligenz in der Unternehmenspraxis. Fraunhofer Verlag, Stuttgart, 2019

[2] Bernijazov R, Dicks A, Dumitrescu R, Foullois M, Hanselle JM, Hüllermeier E, Karakaya G, Ködding P, Lohweg V, Malatyali M, Meyer auf der Heide F, Panzner M, Soltenborn C. A Meta-Review on Artificial Intelligence in Product Creation. Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI 2021). Montreal, Kanada, 2021

[3] Ulich E. Arbeitssysteme als Soziotechnische Systeme – eine Erinnerung. Psychologie des Alltagshandelns, vol. 6, no. 1, 2013, pp. 4-12.

[4] Jarrahi M. Artificial intelligence and the future of work: Human-AI symbiosis in organizational decision making. Business Horizons, Vol. 61, No. 4, 2018, pp. 577-586

[5] Wilson HJ, Daugherty PR. Collaborative Intelligence: Humans and AI Are Joining Forces. Harvard Business Review, 2018

[6] Epstein SL. Wanted: Collaborative intelligence. Elsevier, Artificial Intelligence, Vol. 221, 2015, pp. 36-45

[7] Ransbotham S, Khodabandeh S, Kiron D, Candelon F, Chu M, LaFountain B. Expanding AI's Impact With Organizational Learning. MIT Sloan Management Review and Boston Consulting Group, 2020

[8] Zheng J, Kobayashi Y, Takahashi Y, Yanagida T, Sato T, Hitaka, D. MLCP: A Framework Integrating with Machine Learning and Optimization for Planning and Scheduling in Manufacturing and Services. IEEE 15th International Conference of System of Systems Engineering, 2020, pp. 123-128

[9] Huchler N. Die Rolle des Menschen in der Industrie 4.0 – Technikzentrierter vs. humanzentrierter Ansatz. AIS-Studien, Vol. 9, No. 1, 2016, pp. 57-79

[10] Pantförder D, Schaupp J, Vogel-Heuser B. Making Implicit Knowledge Explicit – Acquisition of Plant Staff's Mental Models as a Basis for Developing a Decision Support System. Communications in Computer and Information Sciences, HIC International 2017 – Posters' Extended Abstracts, 2017, pp. 385-365

[11] DIN Deutsches Institut für Normung e.V. Ergonomie der Mensch-System-Interaktion – Teil 210: Menschenzentrierte Gestaltung interaktiver Systeme (DIN EN ISO 9241-210), 2020

[12] Partsch H. Requirements Engineering systematisch. Springer, 2. Auflage, 2010

[13] Belani H, Vukovic M, Car Z. Requirements Engineering Challenges in Building AI-Based Complex Systems. 27th International Requirements Engineering Conference Workshops (REW), IEEE, 2019, pp. 252-255

[14] Scherf B. Ein Vorgehensmodell zur Personaleinsatzplanung. In: Fank M, Scherf B. (Eds.). Handbuch Personaleinsatzplanung. Datakontext Fachverlag GmbH, Frechen, 2005, pp. 15-52

[15] Burggräf P, Dannapfel M, Esfahani ME, Pulz C, Attig P. Fertigungsmittel- und Personaleinsatzplanung. In: Burggräf P, Schuh G (Eds.). Fabrikplanung – Handbuch Produktion und Management 4, 2. Auflage, Springer Vieweg, Berlin, 2021, pp. 273-364

[16] Mlekus L, Maier GW. More Hype Than Substance? A Meta-Analysis on Job and Task Routine. Frontiers in Psychology, Vol. 12, Article 633530, 2021

[17] Latos B. Auswirkungen von Komplexität auf die Performance von Produktionsgruppen. Schriftenreihe Industrial engineering and ergonomics, Dissertation, RWTH Aachen University, 2020

[18] Schlick C, Bruder R, Luczak H. Arbeitswissenschaft. Springer Vieweg, 4. Auflage, 2018

[19] Paulsen H, Zorn V, Inkermann D, Reining N, Baschin J, Vietor T, Kauffeld S. Soziotechnische Analyse und Gestaltung von Virtualisierungsprozessen. Gruppe. Interaktion. Organisation (GIO), Ausgabe 51, S. 81-93, 2020

[20] Ltifi H, Kolski C, Ayed MB, Alimi AM. A Human-centred Design Approach for Developing Dynamic Decision Support System based on Knowledge Discovery in Databases. Journal of Decision Systems, Vol. 22, No. 2, 2013, pp. 69-96

[21] Bansmann M, Gabriel S, Kühn A, Dumitrescu R. Maturity model-based implementation of scenarios of digitized work. ISPIM Conference Proceeding. The International Society for Professional Innovation Management (ISPIM), 2019

[22] Hobscheidt D, Kühn A, Dumitrescu R. Development of risk-optimized implementation paths for Industry 4.0 based on socio-technial pattern. Procedia CIRP, 2020.

[23] Pitz N, Büttner S, Röcker C. Assistenzsysteme im Kontext von Industrie 4.0 – Partizipative Technologiegestaltung zur Wahrung der Arbeitnehmer*innen-Interessen. Proceeedings of the Mensch und Computer 2020 Workshops, Magdeburg, Deutschland, 2020

[24] Völker M, Carl S, Rose O. Simulationsbasierte Personaleinsatzplanung für komplexe Montagesysteme. Wissenschaftliche Zeitschrift der Technischen Universität Dresden, , Vol. 58, No. 1-2, 2009, pp. 39-44

[25] Spath D, Gerlach S, Hämmerle M, Schlund S, Strölin T. Cyber-Physical System For Self-Organized And Flexible Labour Utilisation. 22nd International Conference on Production Research, 2013

[26] Rischmeyer N. Machine Learning as Key Technology of AI: Automated Workforce Planning. In: Glauner P, Plugmann P, Lerzynski G. Digitalization in Healthcare, 2021, pp. 235-244

[27] Hevner A. A Three Cycle View of Design Science Research. Scandinavian Journal of Information Systems, Vol. 19, No. 2, 2007, pp. 87-92

[28] Gausemeier J, Dumitrescu R, Echterfeld J, Pfänder T, Steffen D, Thielemann F. Innovationen für die Märkte von morgen. Hanser Verlag, München, 2019

[29] DIN Deutsches Institut für Normung e.V. Grundsätze der Ergonomie für die Gestaltung von Arbeitssystemen (ISO 6385:2016), 2016

[30] Gioia DA, Corley KG, Hamilton AL. Seeking Qualitative Rigor in Inductive Research: Notes on the Gioia Methodology. Organizational Research Methods, Vol. 16, No. 1, 2020, pp. 15-31

[31] Bentler D, Gabriel S, Meyer zu Wendischhoff D, Bansmann M, Latos B, Junker C, Maier GW. Gestaltung humanzentrierter Entscheidungen einer künstlichen Intelligenz für Personaleinsatzprozesse produzierender Unternehmen. GfA-Frühjahrskongress, Magdeburg, 2022

This is a resupply of March 2023 as the template used in the publication of the original article contained errors. The content of the article has remained unaffected.